Lecture 6. Offshoring

Lectures on International Trade

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Recall from Previous Lectures

- Trade of final consumption goods and services
- Factor movements across countries
  - Immigration
  - FDI
- How about trade of intermediate parts/tasks?
- Offshoring, or foreign outsourcing
  - The provision of a service or the production of various parts of a good in different countries that are then used or assembled into a final good in another location is called foreign outsourcing or, more simply, offshoring.

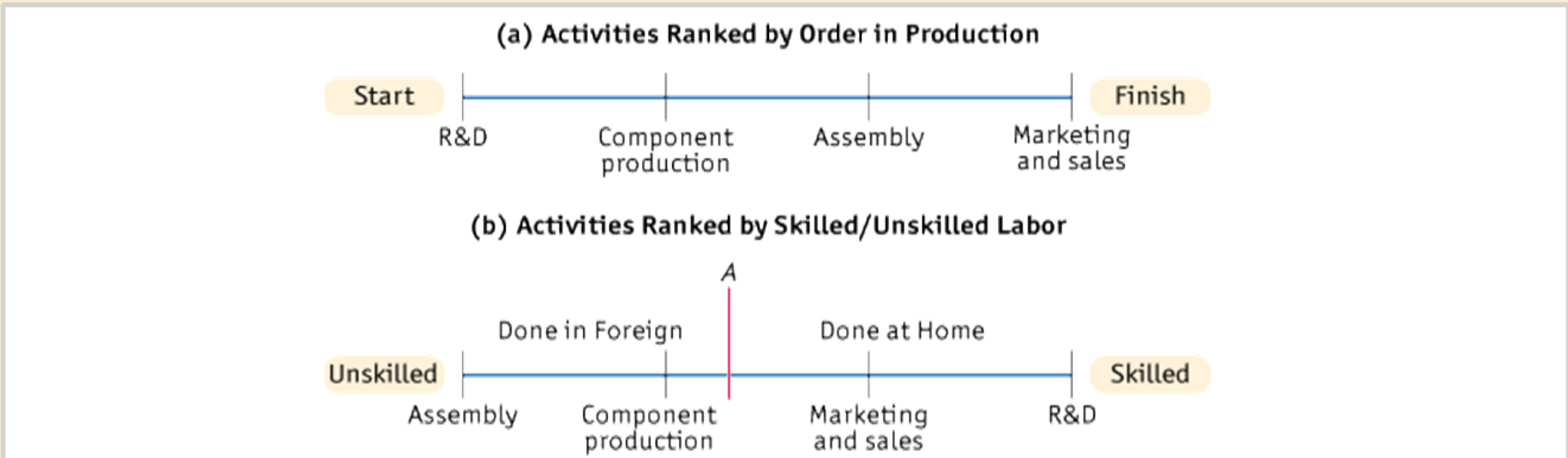
Outline

- A Model of Offshoring
  - The impact of offshoring on the demand and relative wage of workers at home and abroad
  - The gains from offshoring
- Terms of Trade

A Model of Offshoring

A Chain of Production Activities

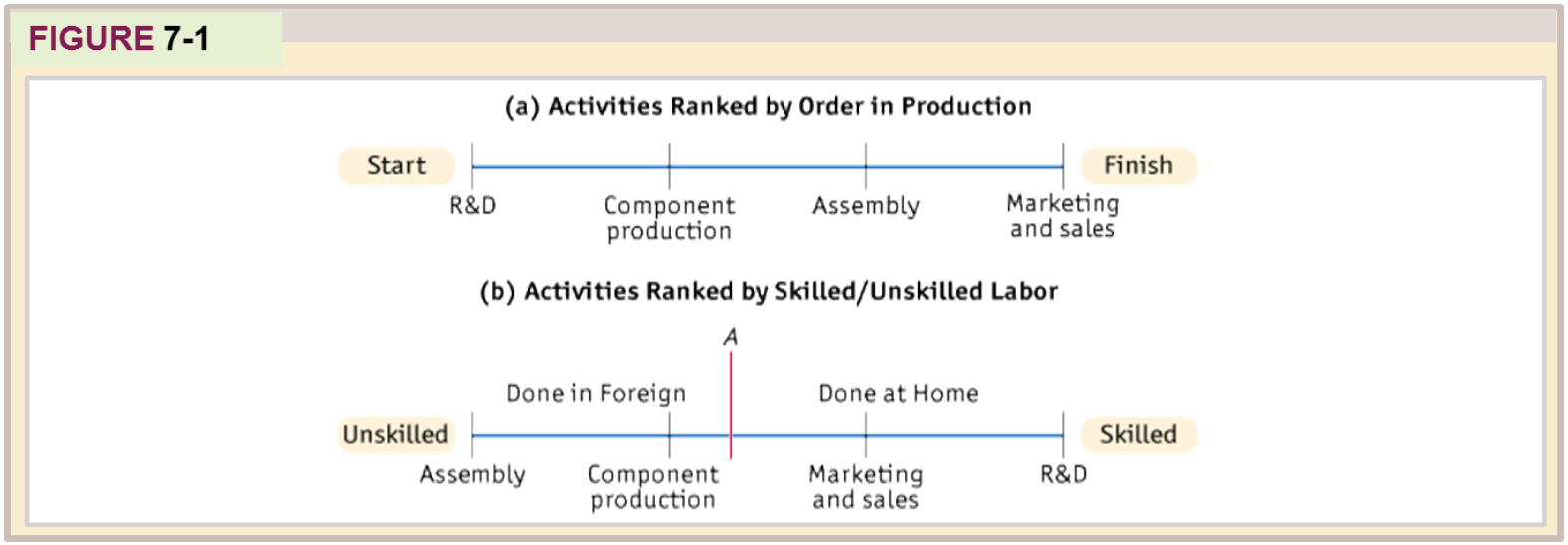
FIGURE 7-1



- Panel (a) lists some of these activities for a given product in the order in which they occur. The value chain in (b) lists these same activities in order of the amount of high-skilled/low-skilled labor used in each.
- Assembly activity, on the left, uses the least skilled labor, and R& D, on the right, uses the most skilled labor.

A Model of Offshoring

A Chain of Production Activities



- If we assume that the relative wage of skilled labor is lower at Home and that trade and capital costs are uniform across activities, there is a point on the value chain, shown by line A, below which all activities are offshored to Foreign and above which all activities are performed at Home.

A Model of Offshoring, cont'

- Our first assumption is that Foreign wages are less than those at Home, so
  - $W_L^* < W_L$ and $W_H^* < W_H$
- Also, that the relative wage of low-skilled labor is lower in Foreign than at Home, so
  - $W_L^*/W_H^* < W_L/W_H$
- Although labor costs are lower in Foreign, the firm must also take into account extra costs of doing business there.
- In making the decision to outsource, a firm will balance the savings from lower wages against the extra costs of capital and trade.
- We assume that these extra costs apply uniformly across all the activities on the value chain-a somewhat unrealistic assumption.

A Model of Offshoring, cont'

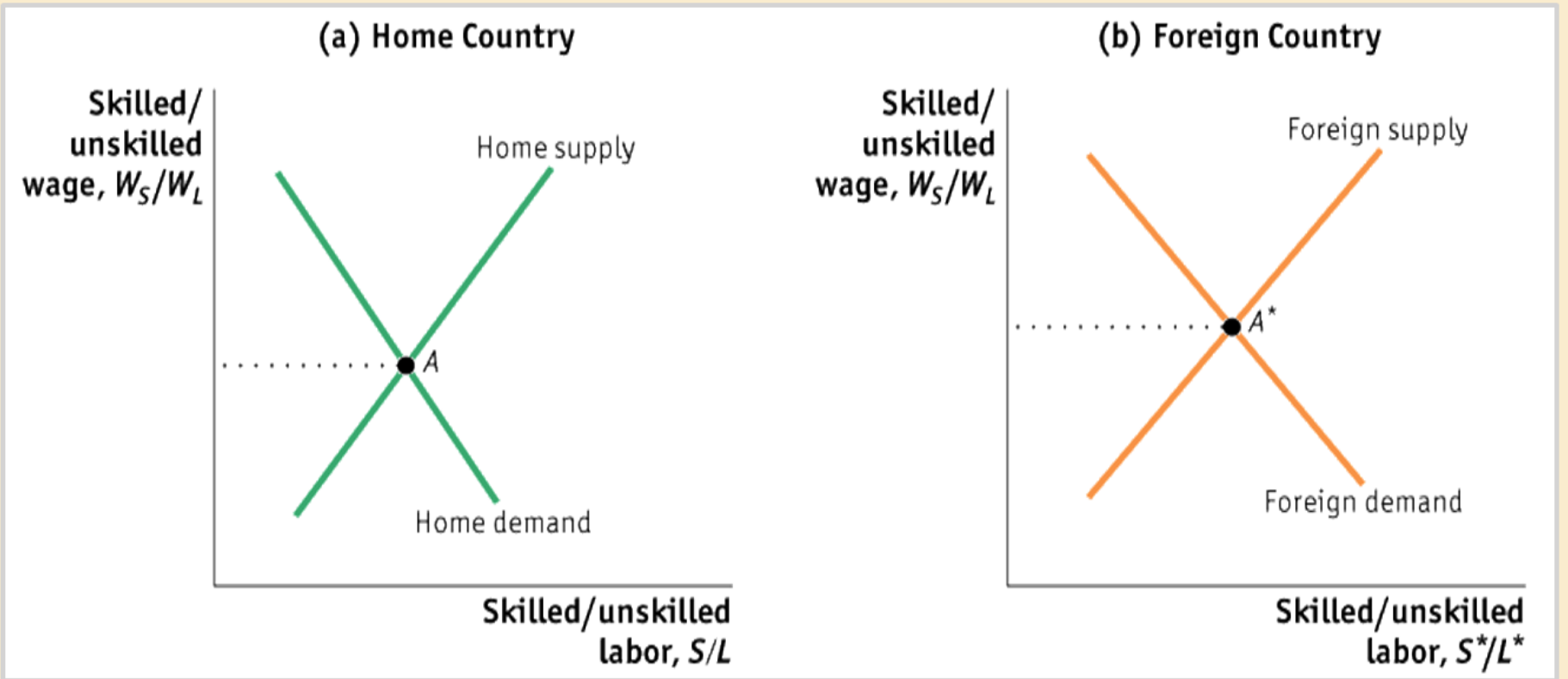
Slicing the Value Chain

- The cost savings on wages are most important in determining which activities to transfer and which to keep at Home.
- This is referred to as *slicing the value chain*.

A Model of Offshoring, cont'

Relative Demand and Supply for Skilled Labor

FIGURE 7-2 **Relative Demand and Supply for Skilled/Unskilled Labor**

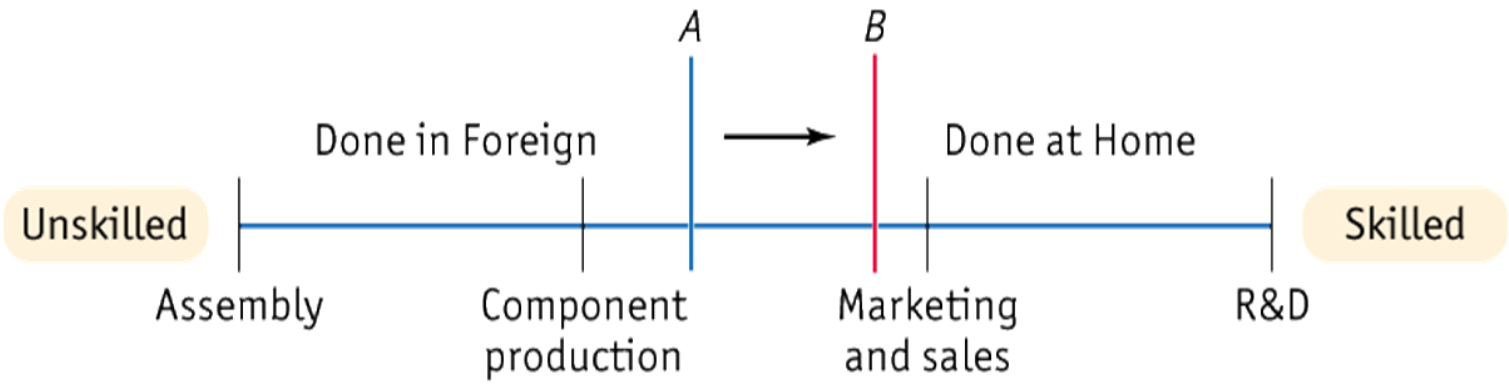


In panel (a), we show the relative demand and supply for skilled labor at Home, H/L, depending on the relative wage, W_H/W_L. In panel (b), we show the relative demand and supply for skilled labor in Foreign, H^*/L^*, depending on the relative wage, W^*_H/W^*_L.

A Model of Offshoring, cont'

Changing the Costs of Trade

FIGURE 7-3



Offshoring on the Value Chain **As the costs of capital or trade fall in the Foreign country, a Home firm will find it profitable to offshore more activities.**

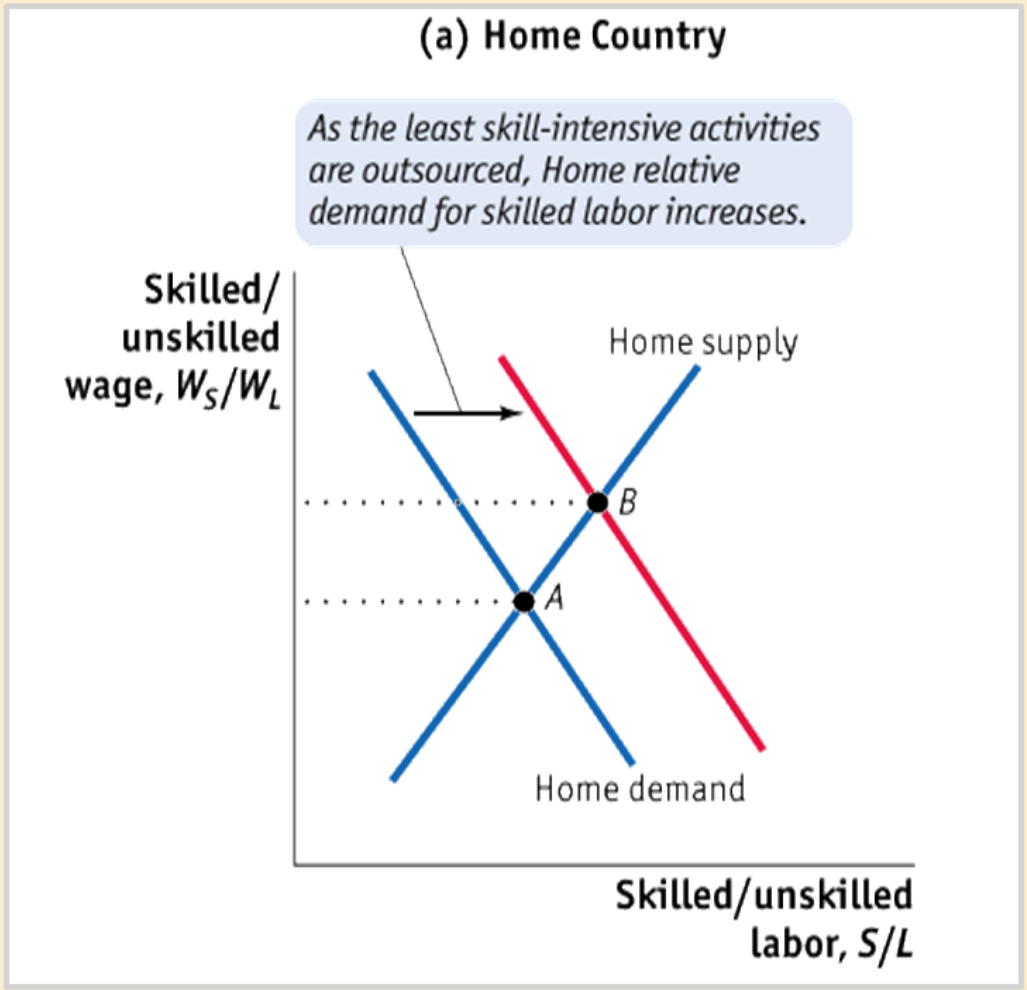
Offshoring shifts the dividing line between Home and Foreign production from *A* to *B*. The activities between *A* and *B*, which formerly were done at Home, are now done in Foreign.

- Notice that these activities are more skill-intensive than the activities formerly done in Foreign (to the left of A) but less skill-intensive than the activities now done at Home (to the right of B).

A Model of Offshoring, cont'

Change in Home Labor Demand and Relative Wage

FIGURE 7-4 (a)



Change in the Relative Demand for High-skilled/Low-skilled Labor

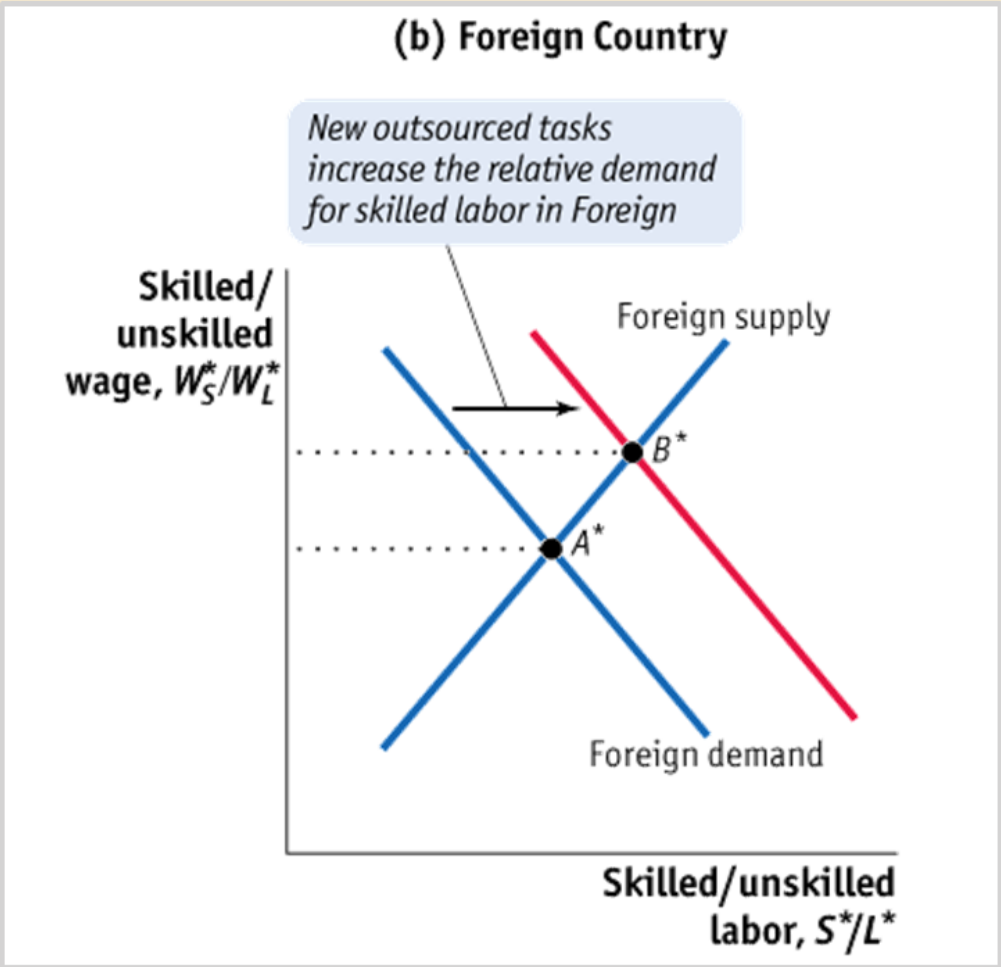
With greater offshoring from Home to Foreign, some of the activities requiring less skill that were formerly done at Home are now done abroad.

It follows that the relative demand for skilled labor at Home increases, and the relative wage rises from point *A* to point *B*.

A Model of Offshoring, cont'

Change in Foreign Labor Demand and Relative Wage

FIGURE 7-4 (b)



Change in the Relative Demand for High-skilled/Low-skilled Labor (continued)

The relative demand for skilled labor in Foreign also increases because the activities shifted to Foreign are more skill intensive than those formerly done there.

It follows that the relative wage for skilled labor in Foreign also rises, from point A^* to point B^*.

A Model of Offshoring, cont'

- In sum, both countries experience an increase in the relative wage of skilled labor due to increased outsourcing.
- As activities in the middle of the value chain are shifted from Home to Foreign, they raise the relative demand for skilled labor in both countries because these activities are the least skill-intensive of those formerly done at Home but the most skill-intensive of those done in Foreign.
- So the relative demand for skilled labor rises in both countries along with the relative wage.
- Can this be used to explain the increasing skill premium?

Gains from Offshoring

- We have shown that outsourcing can shift the relative demand for labor, and raise the wage for skilled workers.
- Since the wage for unskilled workers is the reciprocal of that for skilled workers, outsourcing will decrease the relative wage for unskilled workers.
- However, outsourcing reduces production costs which, in a competitive market, reduces prices. Then outsourcing can have benefits for both producers and their workers, and consumers.
- In this section, we try to balance the potential losses faced by some groups (unskilled labor) with the gains enjoyed by others (skilled labor and consumers).
- In the previous chapters, the Ricardian and Heckscher-Ohlin models generate more gains than losses. Is this true for outsourcing?

Gains from Offshoring, cont'

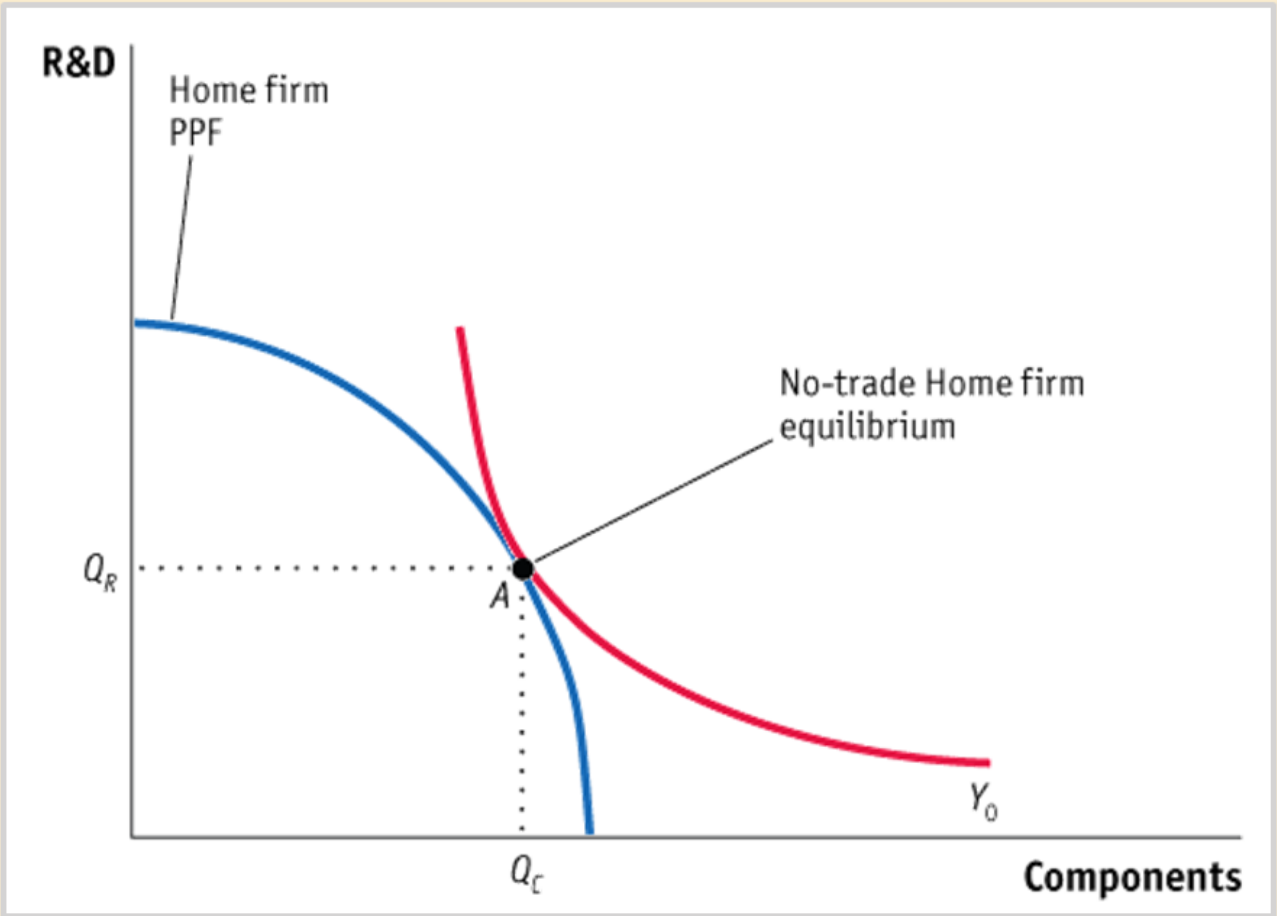
A Simplified Offshoring Model

- Suppose there are only two activities: components production and research and development (R&D).
- Each activity uses skilled and unskilled labor, but we assume that components uses unskilled labor intensively and R&D uses skilled labor intensively.
- The costs of capital are equal in both activities.
- We want to compare the no-trade situation to an equilibrium with trade through outsourcing, to determine if there are overall gains from trade.

Gains from Offshoring, cont'

A Simplified Offshoring Model

FIGURE 7-9 No-Trade Equilibrium for the Home Firm



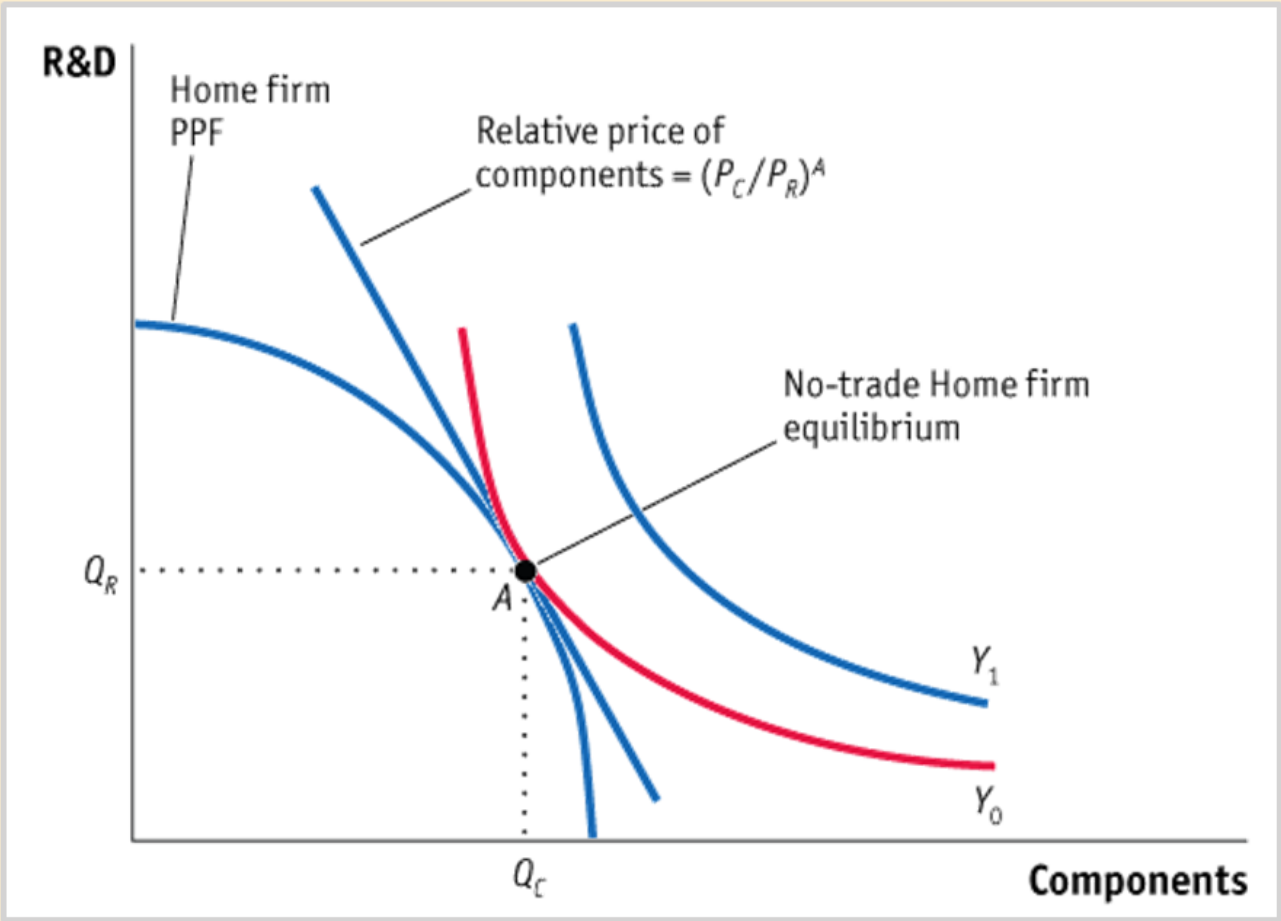
The PPF shows the combinations of components and R&D that can be undertaken by a firm with a given amount of labor and capital.

In the absence of offshoring, the firm produces at *A*, using quantities Q_C of components and Q_R of R&D to produce amount Y_0 of the final good.

Gains from Offshoring, cont'

A Simplified Offshoring Model

FIGURE 7-9 No-Trade Equilibrium for the Home Firm



The line tangent to the isoquant through point *A* measures the value that the firm puts on components relative to R&D, or their relative price, $(P_C/P_R)^A$.

Amount Y_1 of the final good cannot be produced in the absence of offshoring because it lies outside the PPF for the firm.

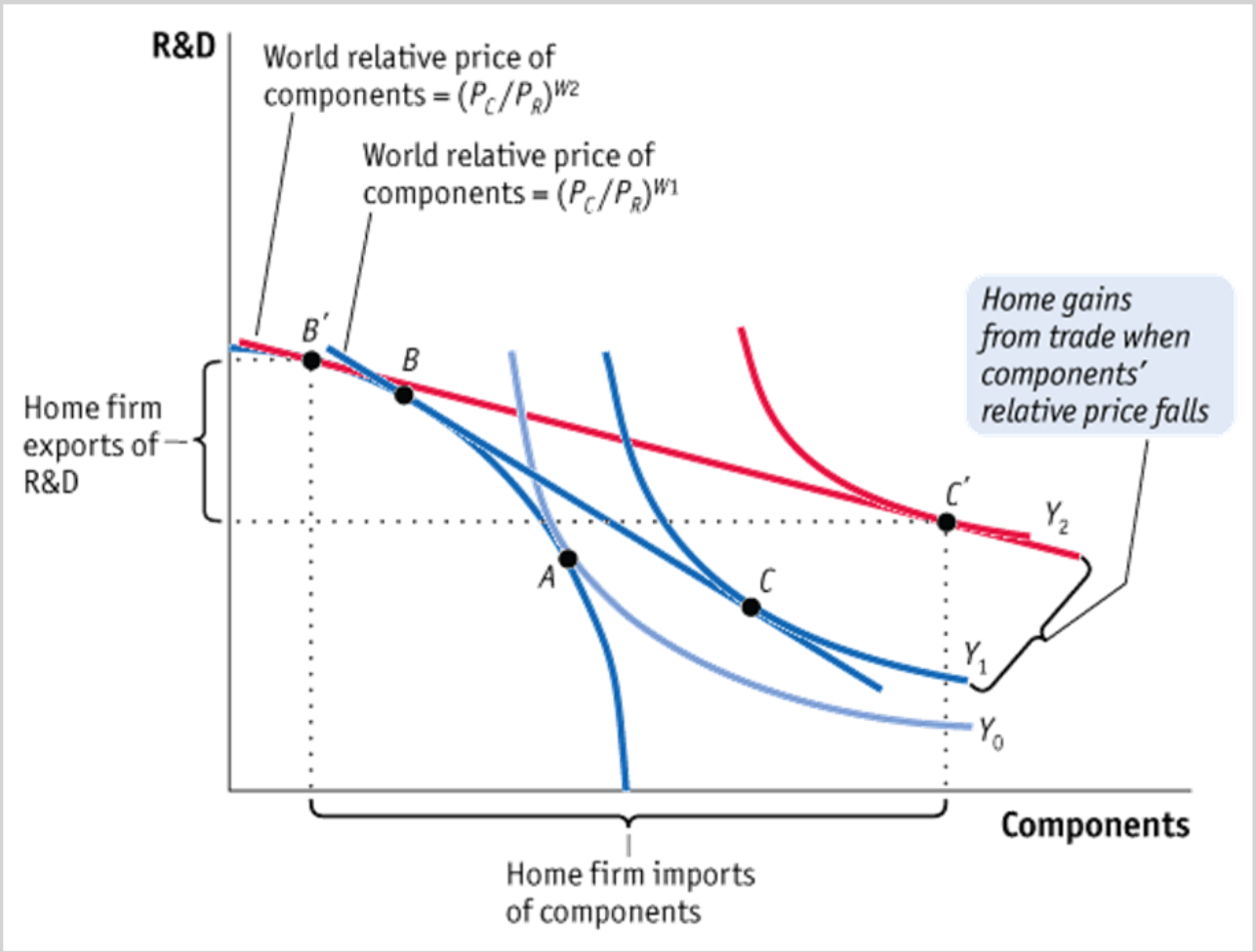
Gains from Offshoring, Equilibrium with Outsourcing

- Suppose the firm can import and export its production activities through outsourcing.
- The quantity of the final good is no longer constrained by the Home PPF.
- A higher level of production (isoquant) is possible by trading intermediate activities.
- We will refer to the relative price of the two activities that the Home firm has available through outsourcing as their world relative price, $(P_C/P_R)^{W1}$.
- Assume the world relative price of components is cheaper than Home's no-trade relative price.
- With a lower relative wage of unskilled labor in Foreign, the components assembly will also be cheaper in Foreign.
- It follows that Home will want to outsource components, which are cheaper abroad, while Home firms will be exporting R&D, which is cheaper at Home.

Gains from Offshoring, cont'

Equilibrium with Outsourcing

FIGURE 7-11 **Fall in the Price of Components**



If the relative price of components falls from $(P_C/P_R)^{W1}$ to $(P_C/P_R)^{W2}$, then the Home firm will do even more R&D and less components production, at point B' rather than B.

The increase in the terms of trade allows the Home firm to produce output Y_2 at point C', and the gains from trade are higher than in the initial offshoring equilibrium (points B and C).

Gains from Offshoring, cont'

Gains from Offshoring within the Firm

- The increase of final goods produced (Y0 – Y1) is a measure of the gains from trade to the Home firm from outsourcing.
- Because more of the final good is produced with the same overall amount of skilled and unskilled labor available in Home, the Home company is more productive. Both its costs of production, and the price of the final product falls.
- The gains for this company are also spread to consumers.
- Therefore: *When comparing a no-trade situation to the equilibrium with offshoring, and assuming that the world relative price differs from that at Home, there are always gains from offshoring*.

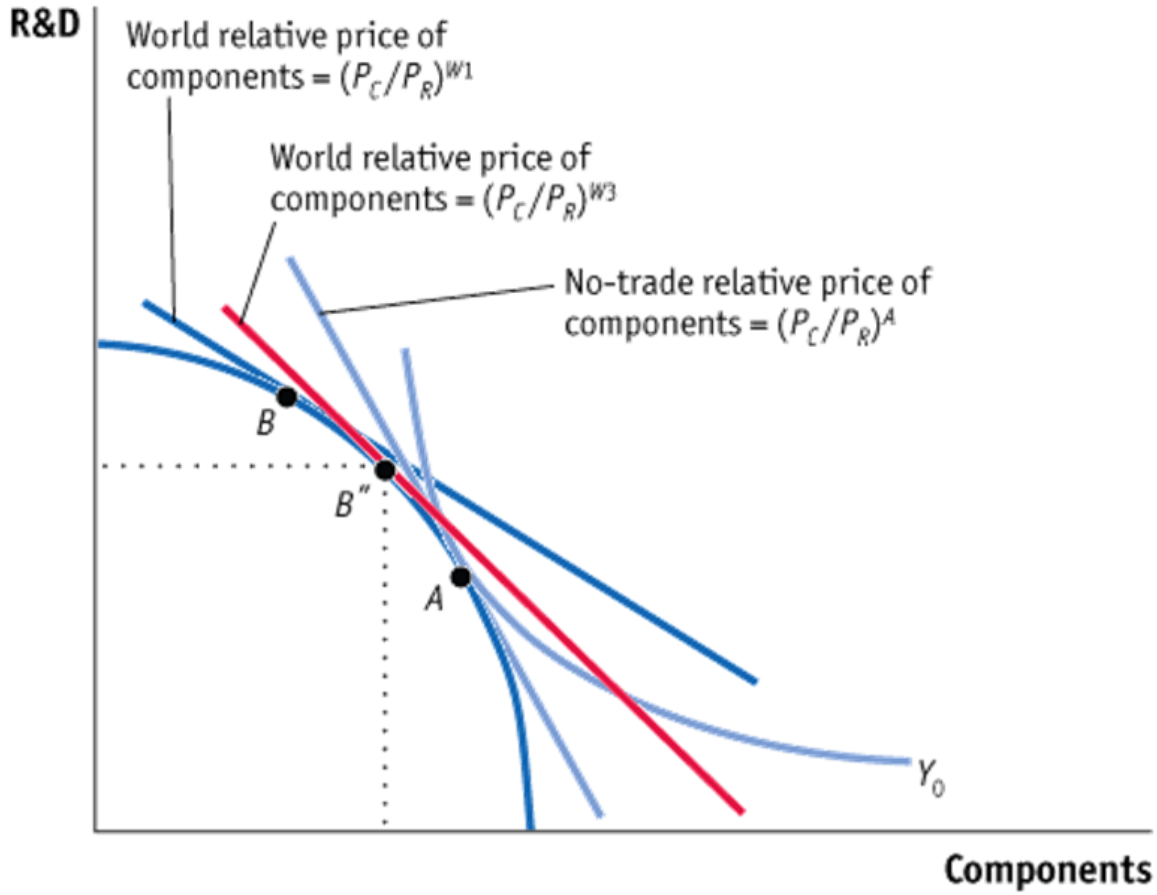
Terms of Trade

- A country's **terms of trade** equal the price of the country's exports divided by the price of its imports. The home terms of trade are $(P_R/P_C)^{W1}$.
- Home is exporting R&D and importing components.
- A rise in the terms of trade indicates that a country is getting a higher price for its exports, or paying a lower price for its imports-both benefit the country.
- Conversely, a fall in the terms of trade harms a country, because it is paying more for its imports and getting less for its exports.

Terms of Trade, cont'

Fall in the price of R&D

FIGURE 7-12 (1 of 3) **A Fall in the Price of R&D**



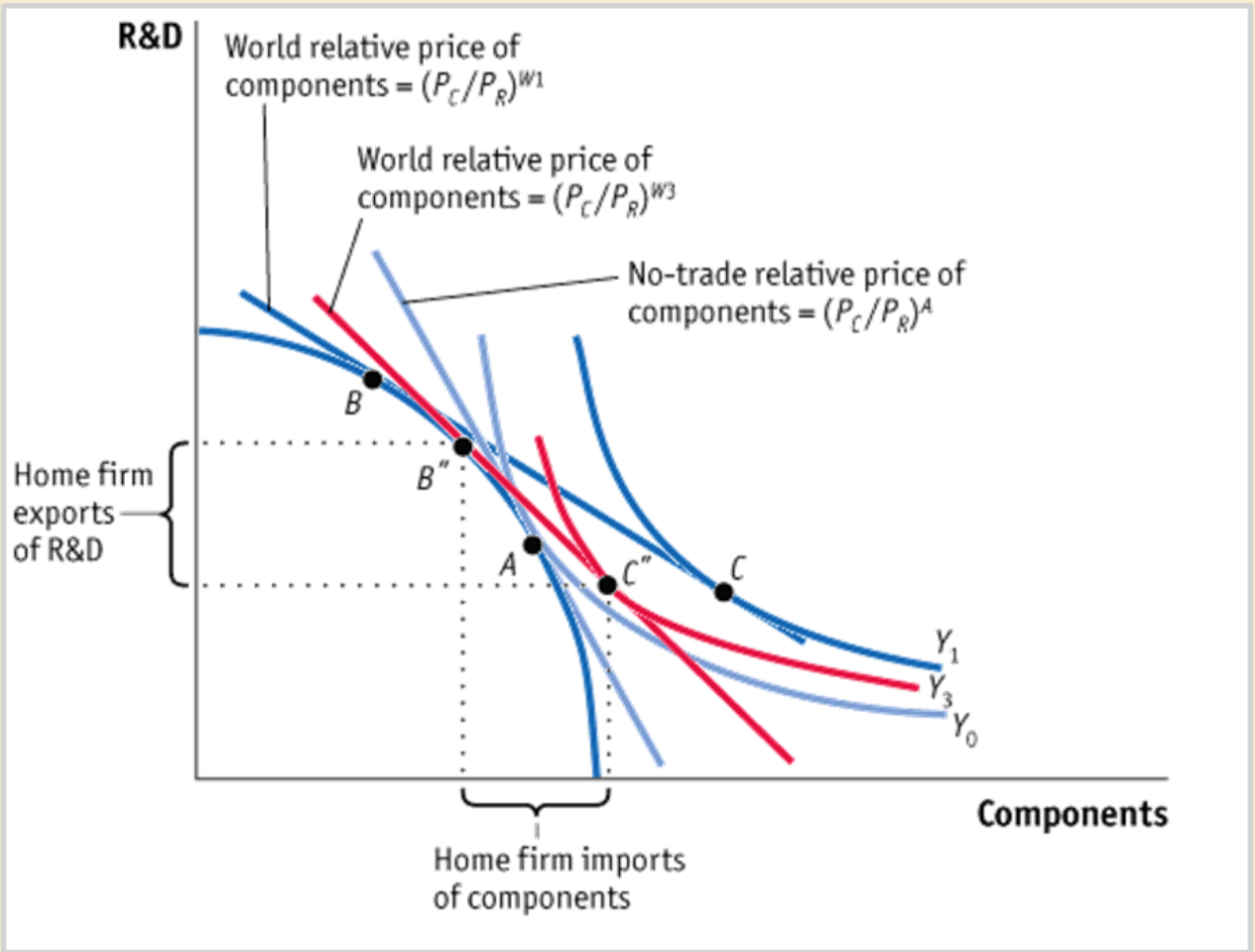
A fall in the relative price of R&D makes the world price line steeper $(P_C/P_R)^{W3}$.

As a result, the Home firm reduces its R&D activities and increases its components activities, moving from *B* to *B″* along the PPF.

Terms of Trade, cont'

Fall in the price of R&D

FIGURE 7-12 (2 of 3) **A Fall in the Price of R&D (continued)**

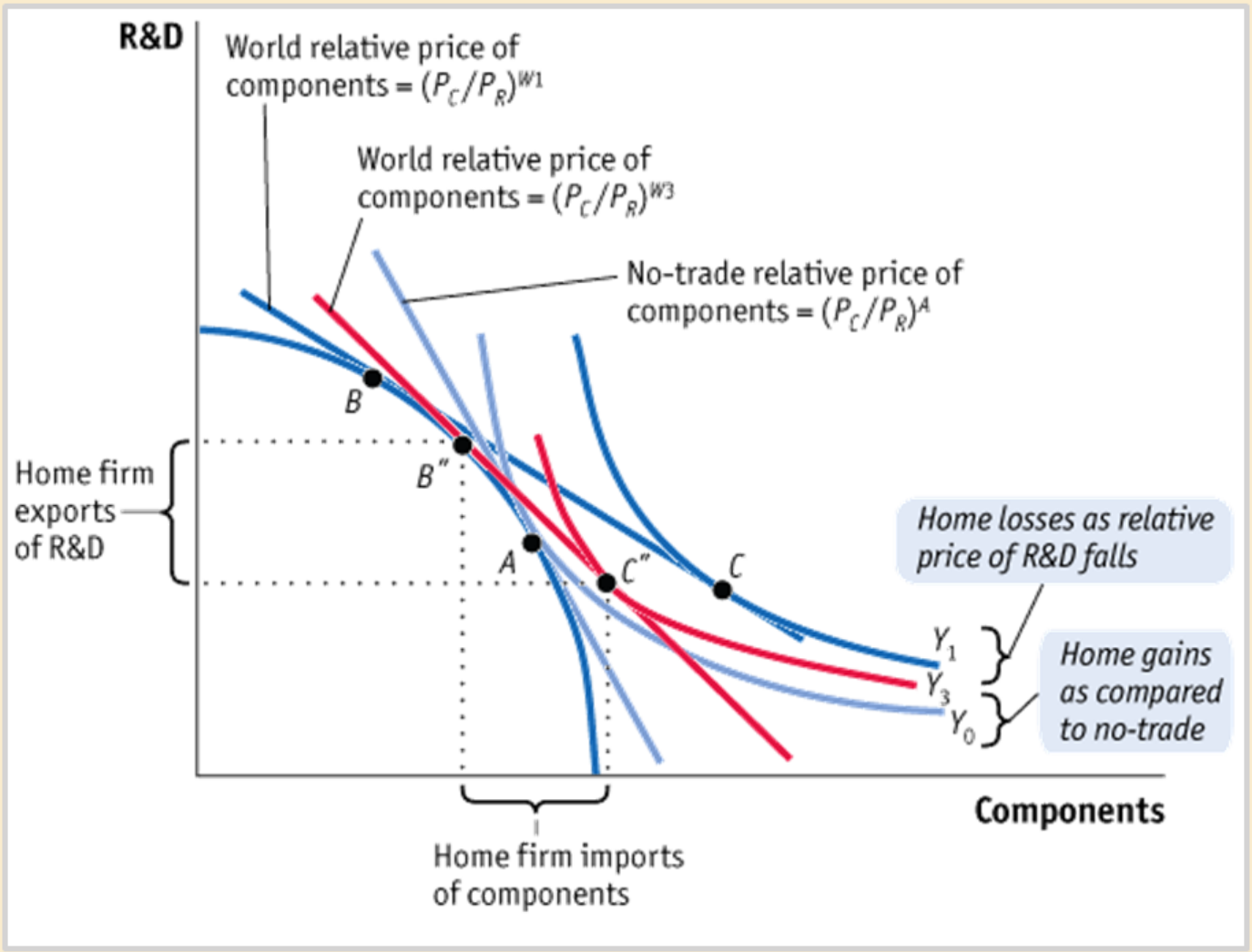


At the new world relative price, the Home firm faces a terms-of-trade loss and can no longer export each unit of R&D for as many components as it could in the initial offshoring equilibrium. The final good output is reduced from Y_1 to Y_3 at point C''. Notice that the final good output Y_3 is still higher than output without trade, Y_0.

Terms of Trade, cont'

Fall in the price of R&D

FIGURE 7-12 (3 of 3) **A Fall in the Price of R&D (continued)**



After the fall in the relative price of R&D, there are still gains from trade relative to no-trade (point *A*).

But losses relative to the initial offshoring equilibrium (points *B* and *C*).

Offshoring

- Today, the focus is frequently on of the newer forms of offshoring, the offshoring of business services to foreign countries.
- Business services are activities such as accounting, auditing, human resources, order processing, telemarketing, and after-sales service, like getting help with your computer.
- Examples: customer service in India; AIG; Consulting firms; etc.

Summary

Key Points:

1. The provision of a service or the production of various parts of a good in different countries for assembly into a final good in another location is called foreign outsourcing or offshoring.

2. We can apply the same ideas that we developed for trade in final goods among countries to the trade of intermediate offshored activities. For instance, if low-skilled labor is relatively inexpensive in the Foreign country, then the activities that are least skill-intensive will be offshored there, and Home will engage in the activities that are more skill-intensive.

Summary, cont'

3. We can also predict what happens to relative wages of skilled labor when there is a change in trading costs and more offshoring. Our model predicts that the relative demand for skilled labor increases in both countries. This result helps to explain the observation that relative wages have been increasing in the United States and in other countries at the same time.

4. In an overall sense, there are gains from offshoring, because the specialization of countries in different production activities allows firms in both countries to produce a higher level of final goods. That increase in output represents a productivity gain, and the gains from trade.

Summary, cont'

5. With service offshoring, it is possible that a country like India will have rising productivity in activities in which the United States has comparative advantage, such as R&D. Rising productivity in India would lead to a fall in the price of R&D, which is a terms-of trade loss for the United States. For that reason, the United States could lose due to service offshoring, though it still gains as compared with a situation of no offshoring at all.